

Business assumptions

Average purchase value (\$)
15.00

Cost of sending the better coupon (\$)
2.00

Targeting rule
☒ Value threshold
☐ Budget cap

Filter held-out customers

Coupon type

Bar x

Carry out & Take a... x

Coffee House x

Restaurant(20-50) x

Restaurant(<20) x

Destination

Home x

No Urgent Place x

Work x

Income bracket

\$100000 or More x

\$12500 - \$24999 x

\$25000 - \$37499 x

\$37500 - \$49999 x

\$50000 - \$62499 x

\$62500 - \$74999 x

\$75000 - \$87499 x

\$87500 - \$99999 x

Less than \$12500 x

Causal ML for Coupon Targeting

Which customers should receive a 1-day coupon instead of a 2-hour coupon? Every reported causal estimate and targeting result below is computed on a held-out evaluation set, not the rows used to train the models.

Overview & Diagnostics Top Customers Customer Lookup

Showing 3,153 filtered customers from a 3,153-row held-out evaluation set.

Customers to target	Incremental revenue	Campaign cost	Campaign ROI
1,832	\$5,433	\$3,664	48.3%

Held-out causal estimates

T-learner average uplift	Propensity matching ATT	Doubly-robust AIPW ATE
+0.148	+0.171	+0.170
↑ 95% CI [+0.129, +0.159]	↑ 95% CI [+0.120, +0.229]	↑ 95% CI [+0.132, +0.200]

The estimators use different assumptions. Agreement is reassuring, but it does not by itself prove that all confounding has been removed.

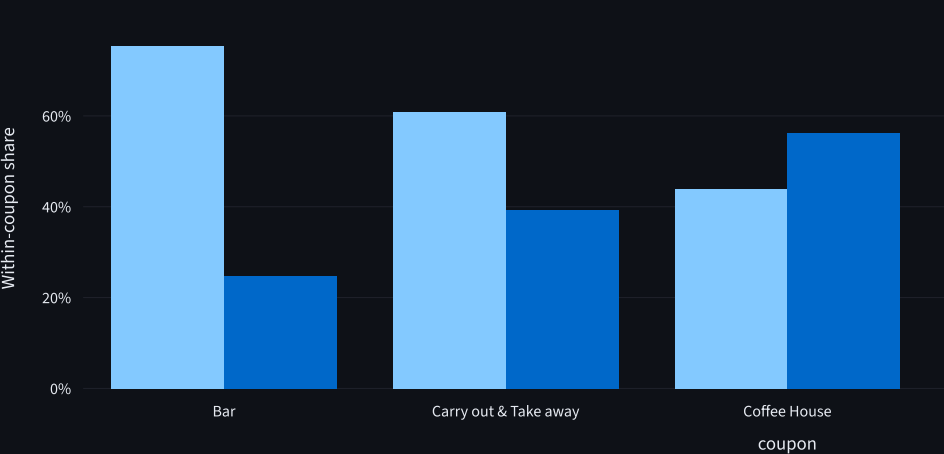
Positivity, overlap, and treatment-balance checks

Control share (2h)	Treated share (1d)	Propensities clipped
44.2%	55.8%	1 (0.0%)

No coupon type exceeds 90% concentration in a single expiration arm.

Some held-out rows reached the 0.01/0.99 propensity clipping bounds and therefore have weak empirical counterfactual support.

Expiration mix within each coupon type



Propensity-score matching quality

PSM match rate	Treated unmatched	Max SMD before	Max SMD after
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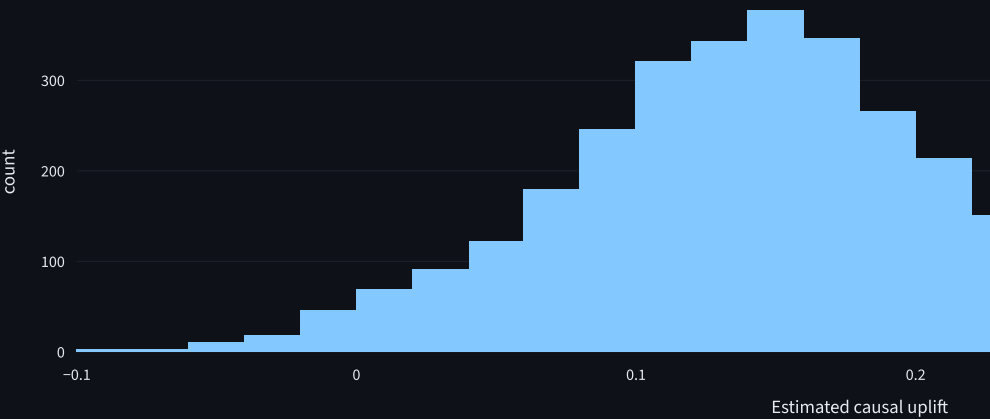
96.0%710.3310.237

Caliper: 0.20 SD of logit propensity score (width=0.185).

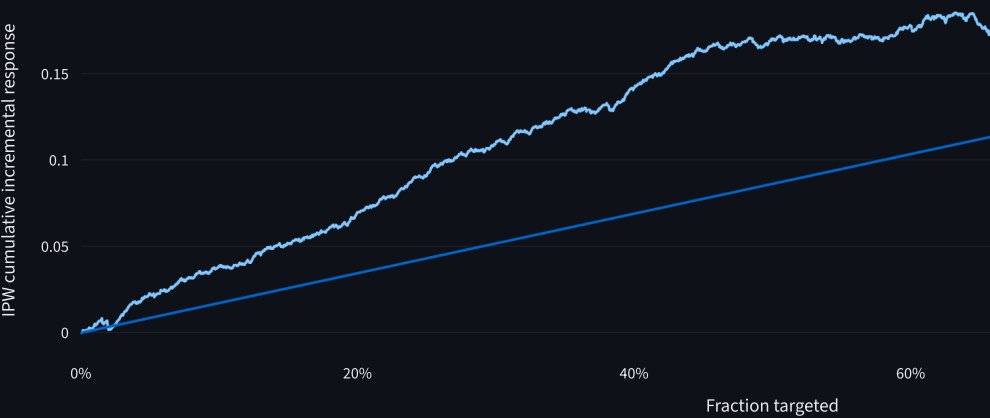
11 features still have |SMD| > 0.10 after matching.

	feature	smd_before	smd_after
18	time_10PM	-0.126	
15	passanger_Partner	-0.05	
3	direction_same	-0.098	
19	time_2PM	0.226	
12	destination_Work	0.105	
21	time_7AM	0.105	
28	maritalStatus_Single	0.065	
16	weather_Snowy	0.194	
24	coupon_Restaurant(20-50)	0.113	
25	coupon_Restaurant(<20)	-0.129	

Held-out distribution of estimated individual uplift



Held-out uplift/Qini-style cumulative gain



Held-out average uplift by coupon type

